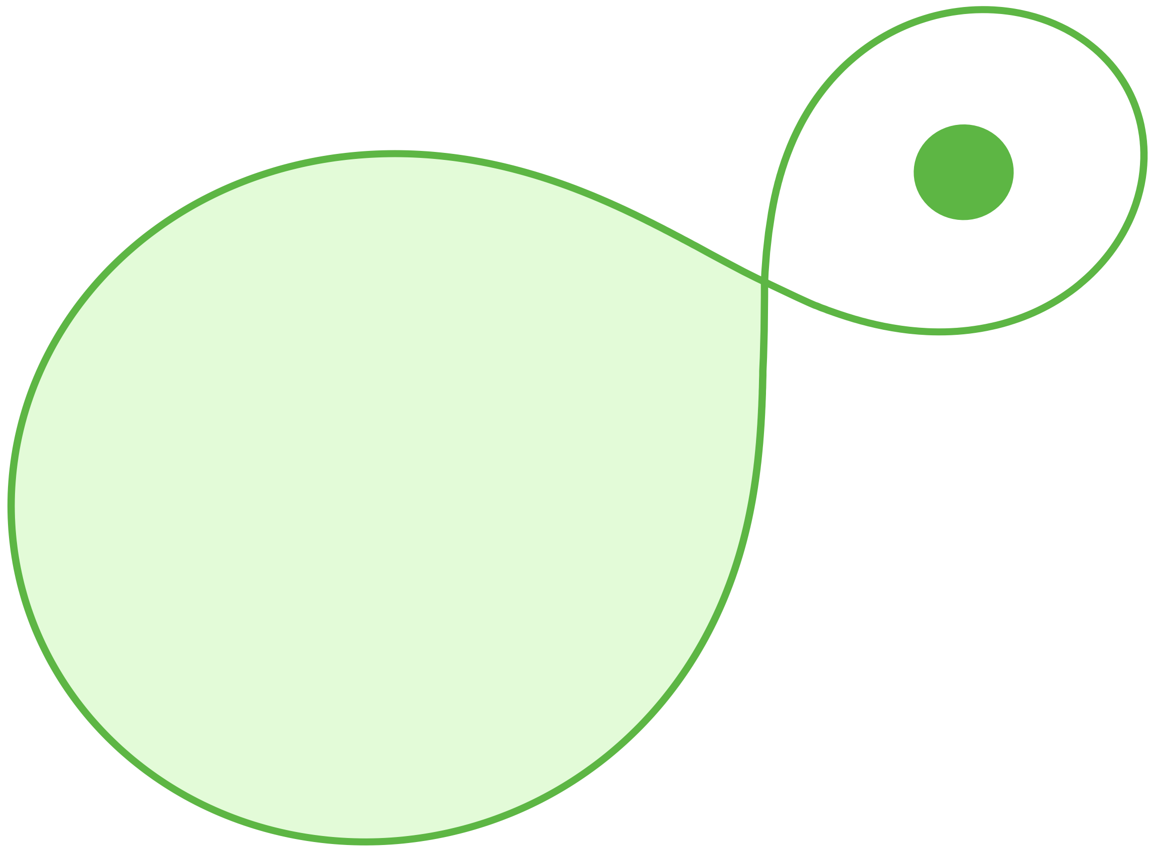
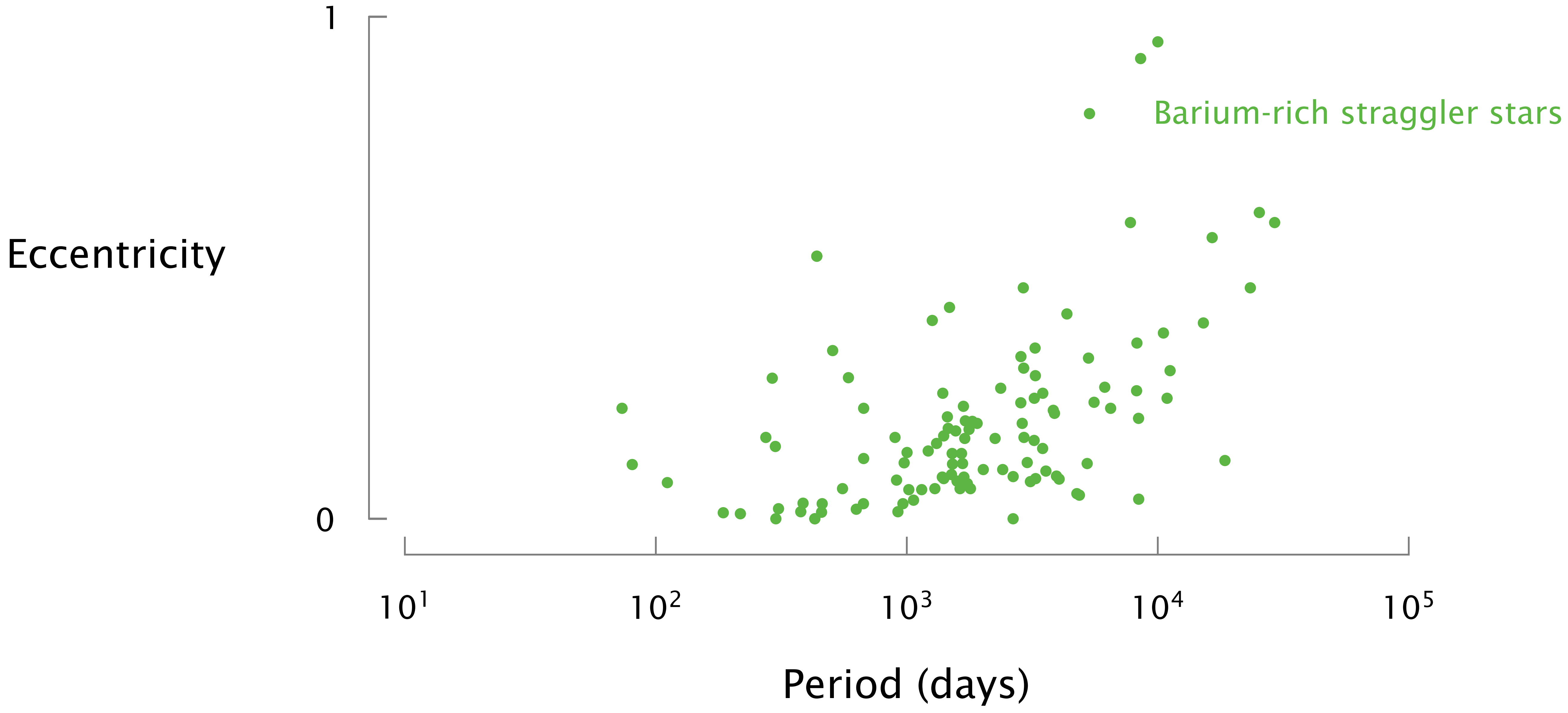
# Forming barium stars through stable RLOF mass transfer from TPAGB-stars

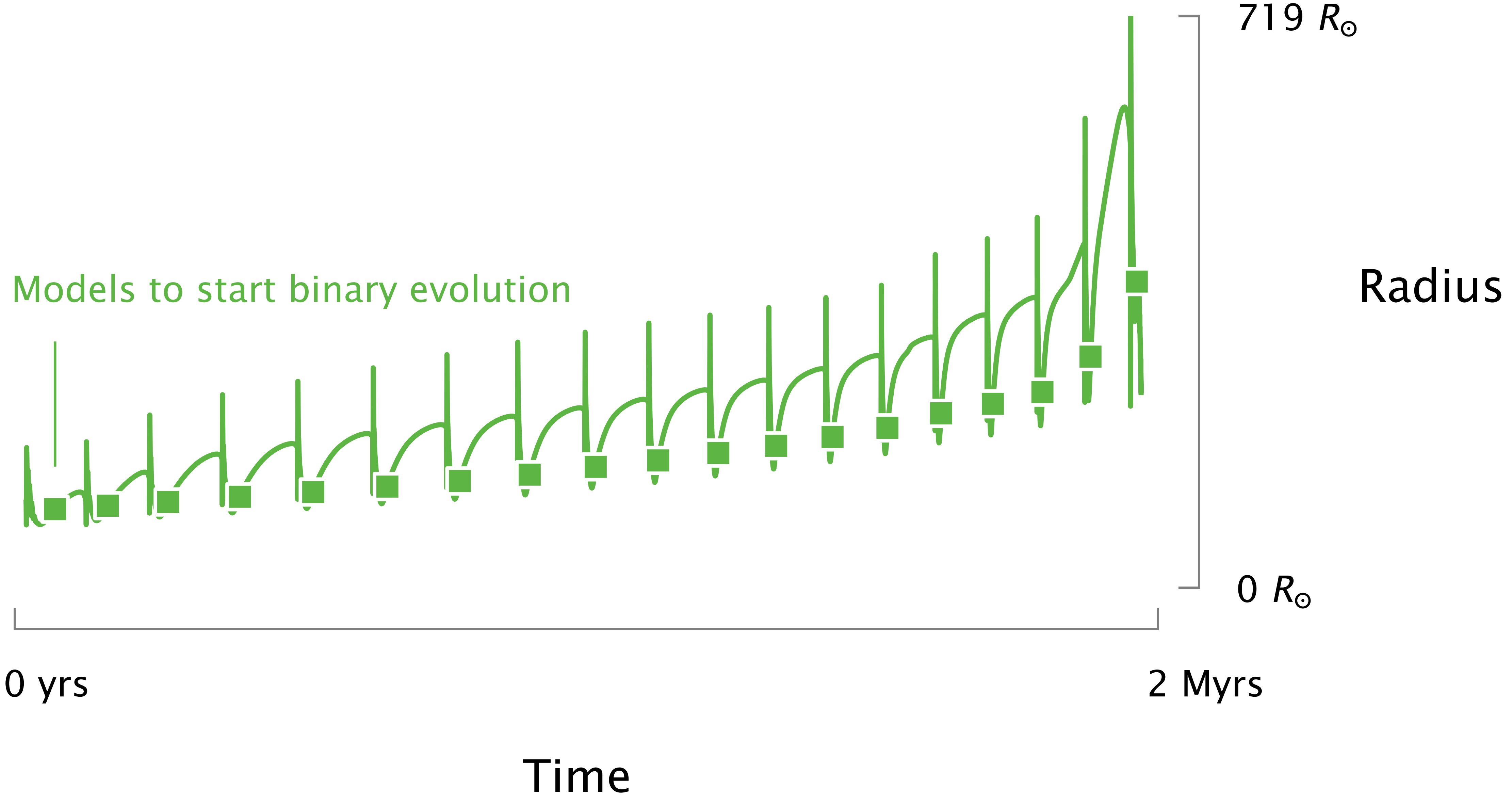


Koen Essers
Radboud University
10$^{th}$ of June 2026

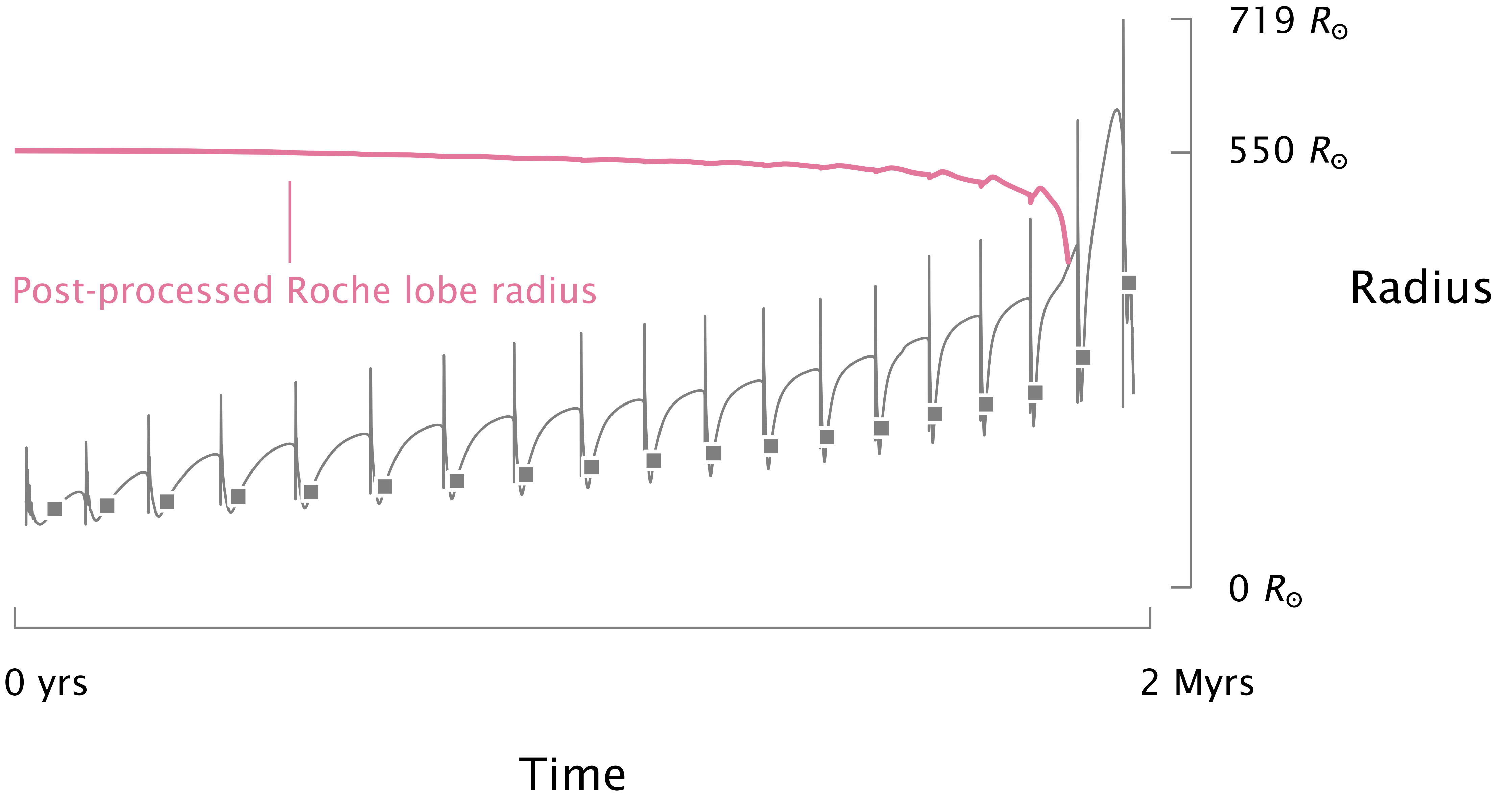
# Ba stars have periods incompatible with wind mass transfer or common envelopes



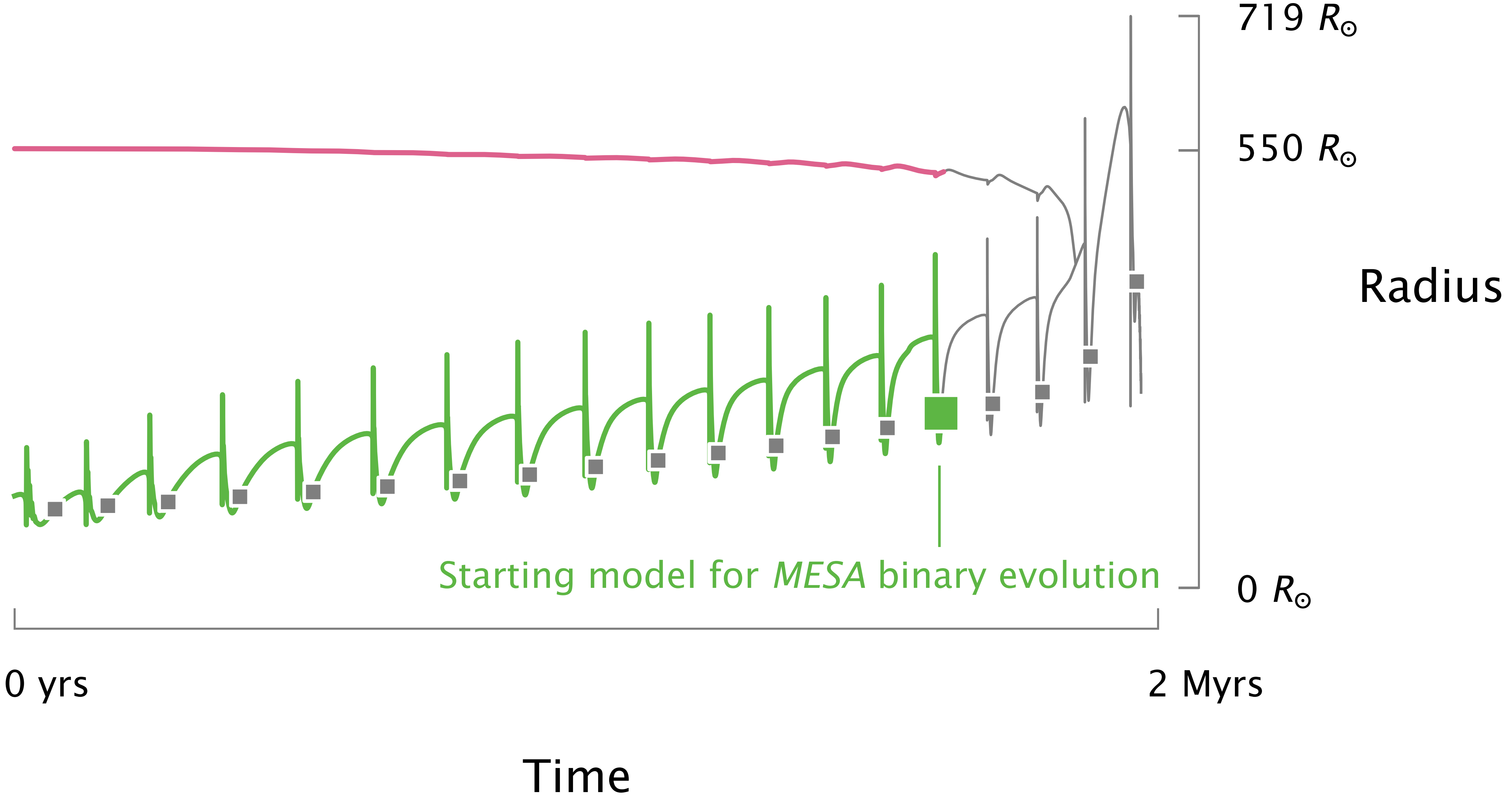
# I simulate detailed TP-AGB stars with *MESA* and I save models to use as binary input



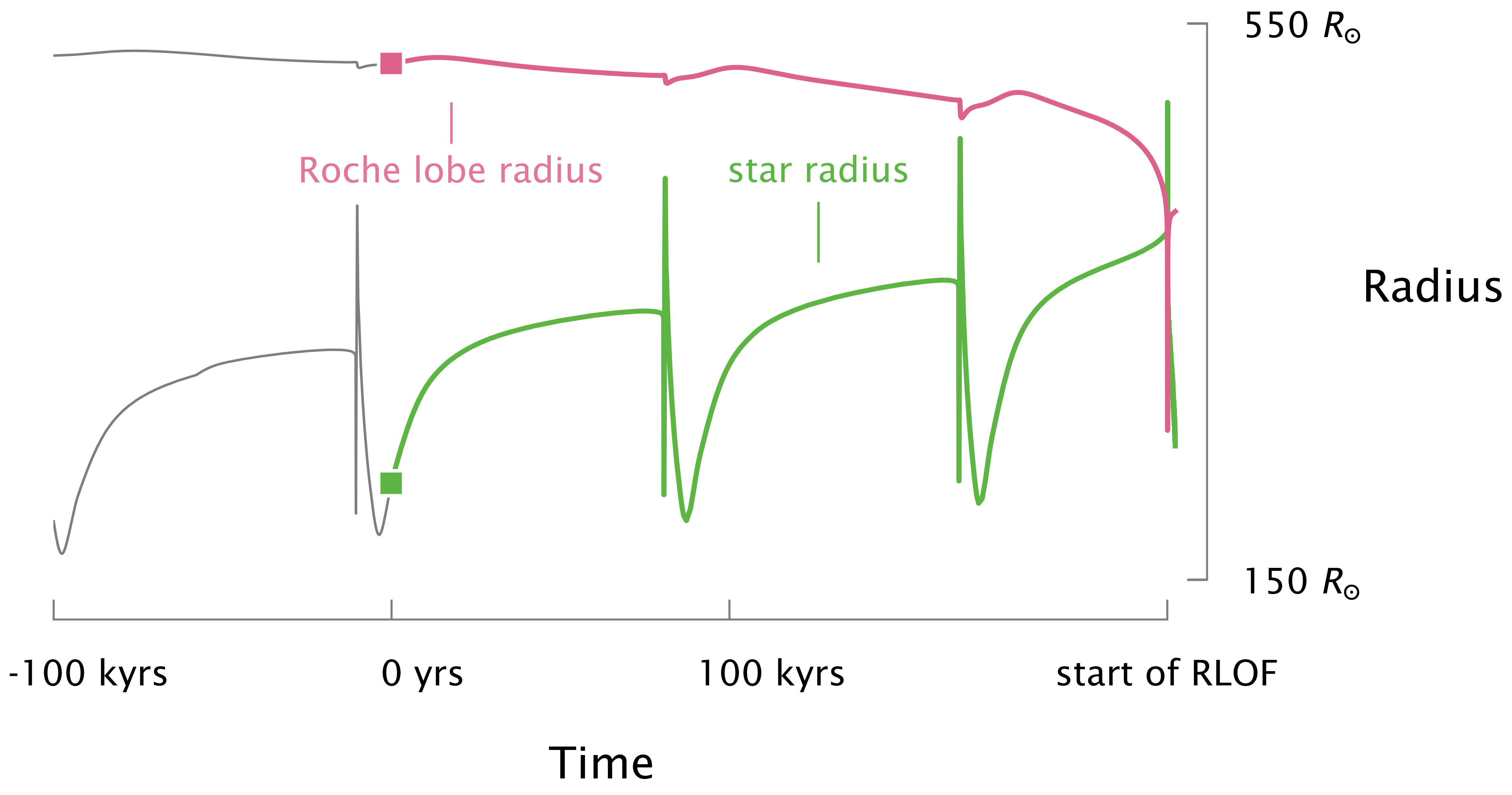
# I predict how the binary evolves using single star models and a post-processing step



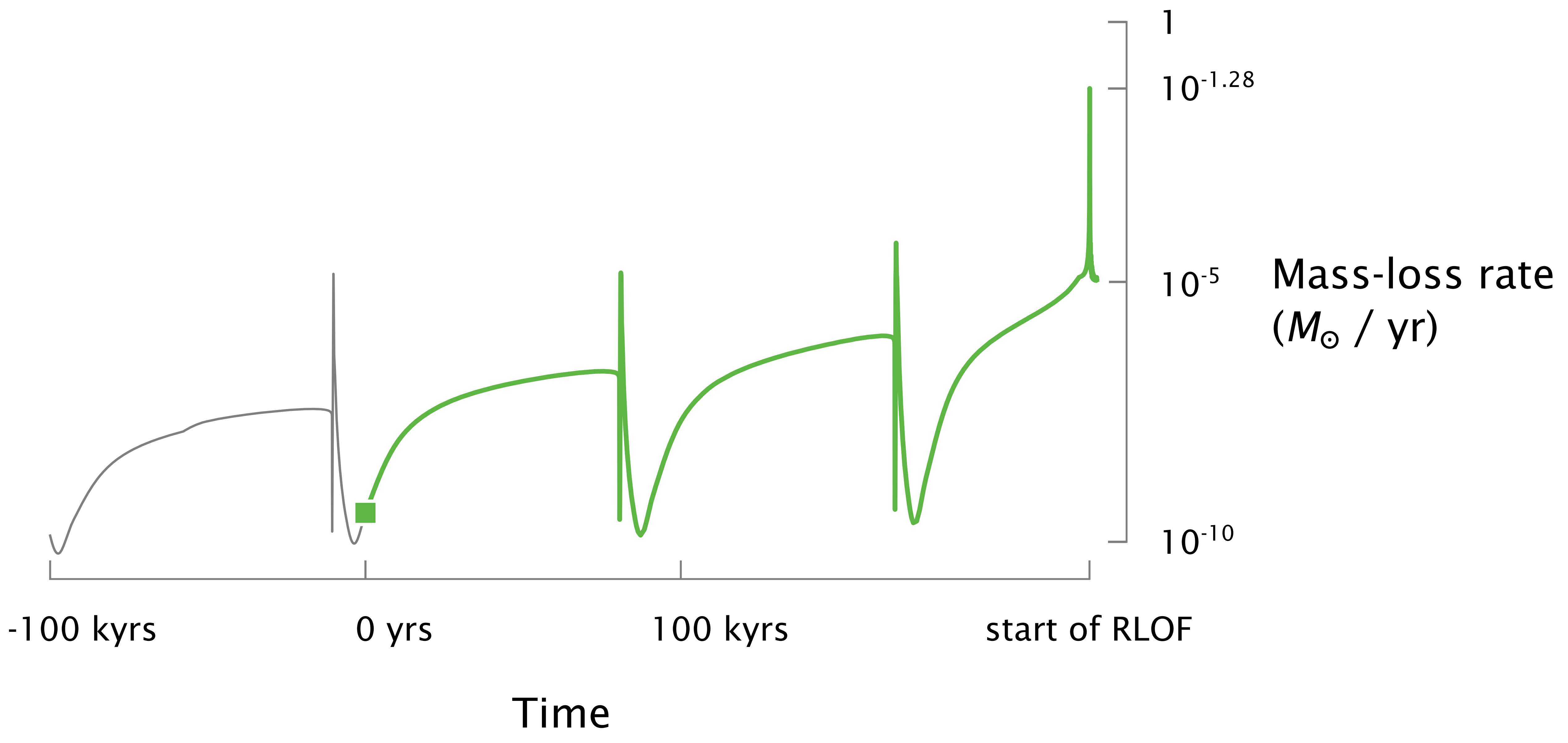
# I determine a *MESA* starting model and infer the rotation, separation and mass-ratio



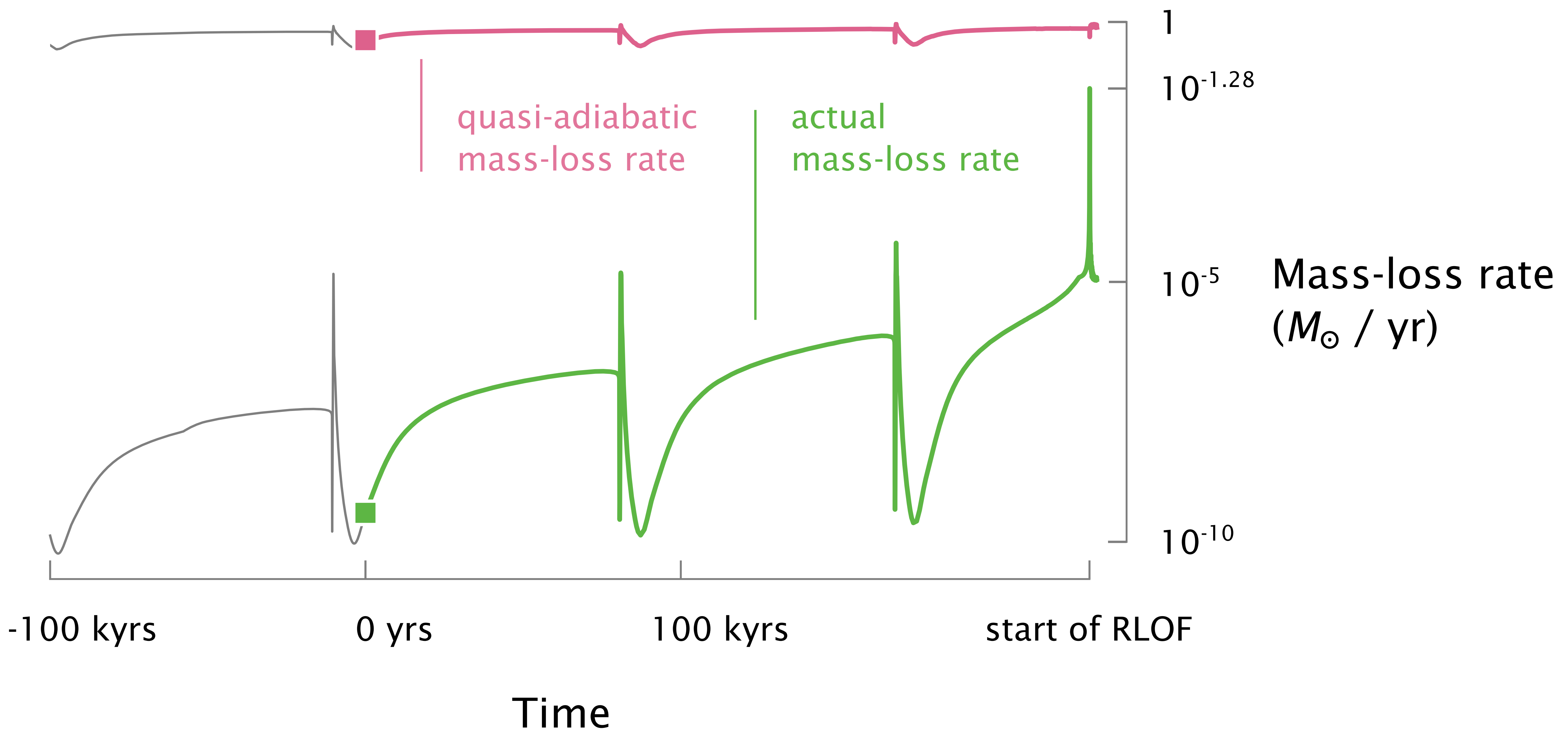
# Tides, wind mass-transfer and stellar evolution drive systems into RLOF



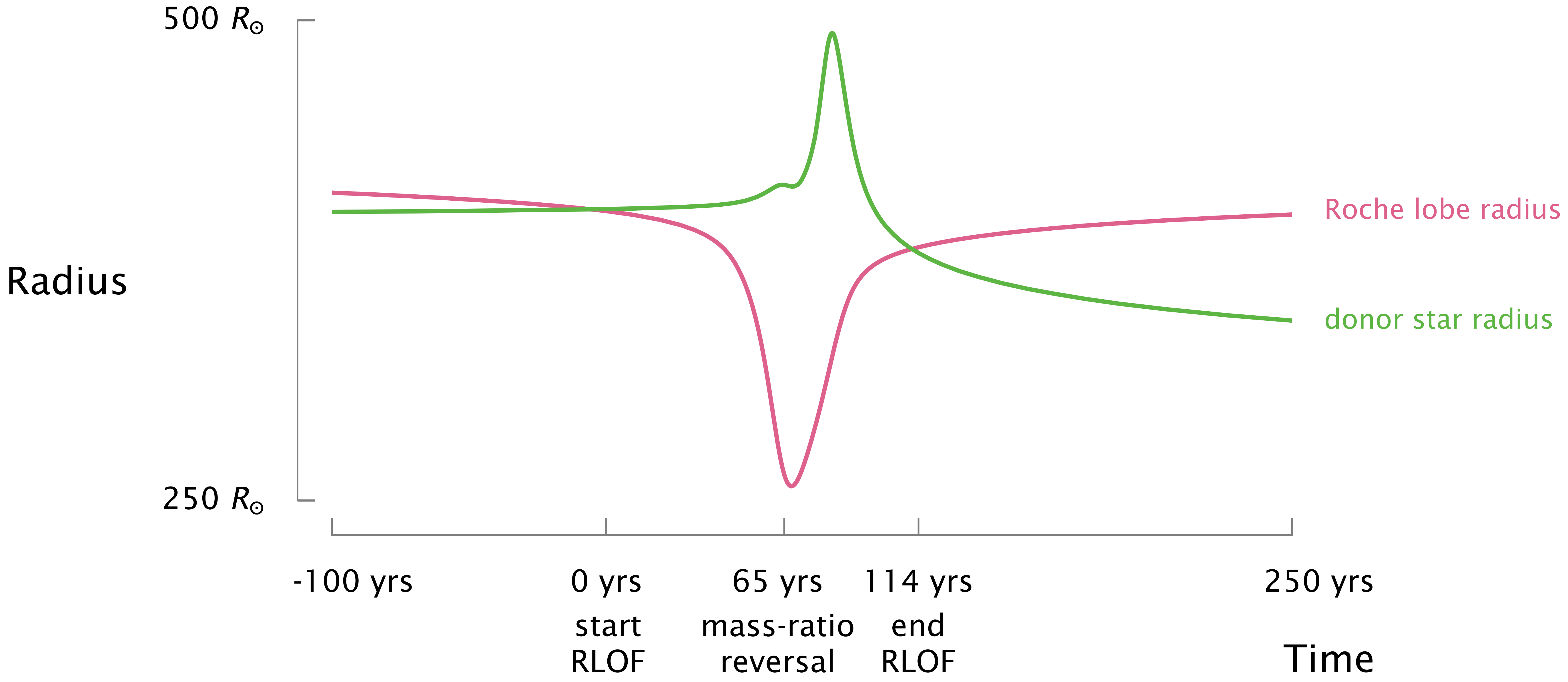
# The donor experiences a short but strong burst of mass loss during RLOF



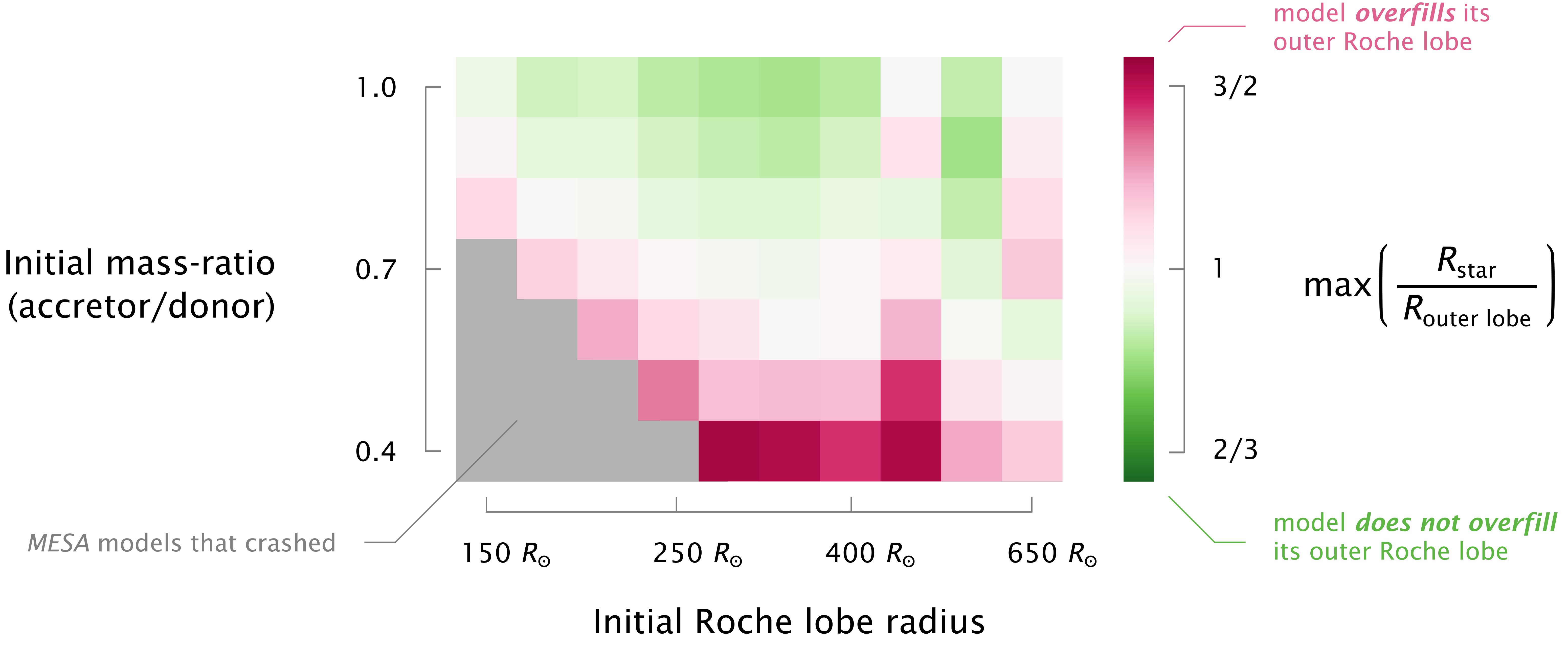
# The quasi-adiabatic mass-loss rate is always larger than the actual mass loss rate

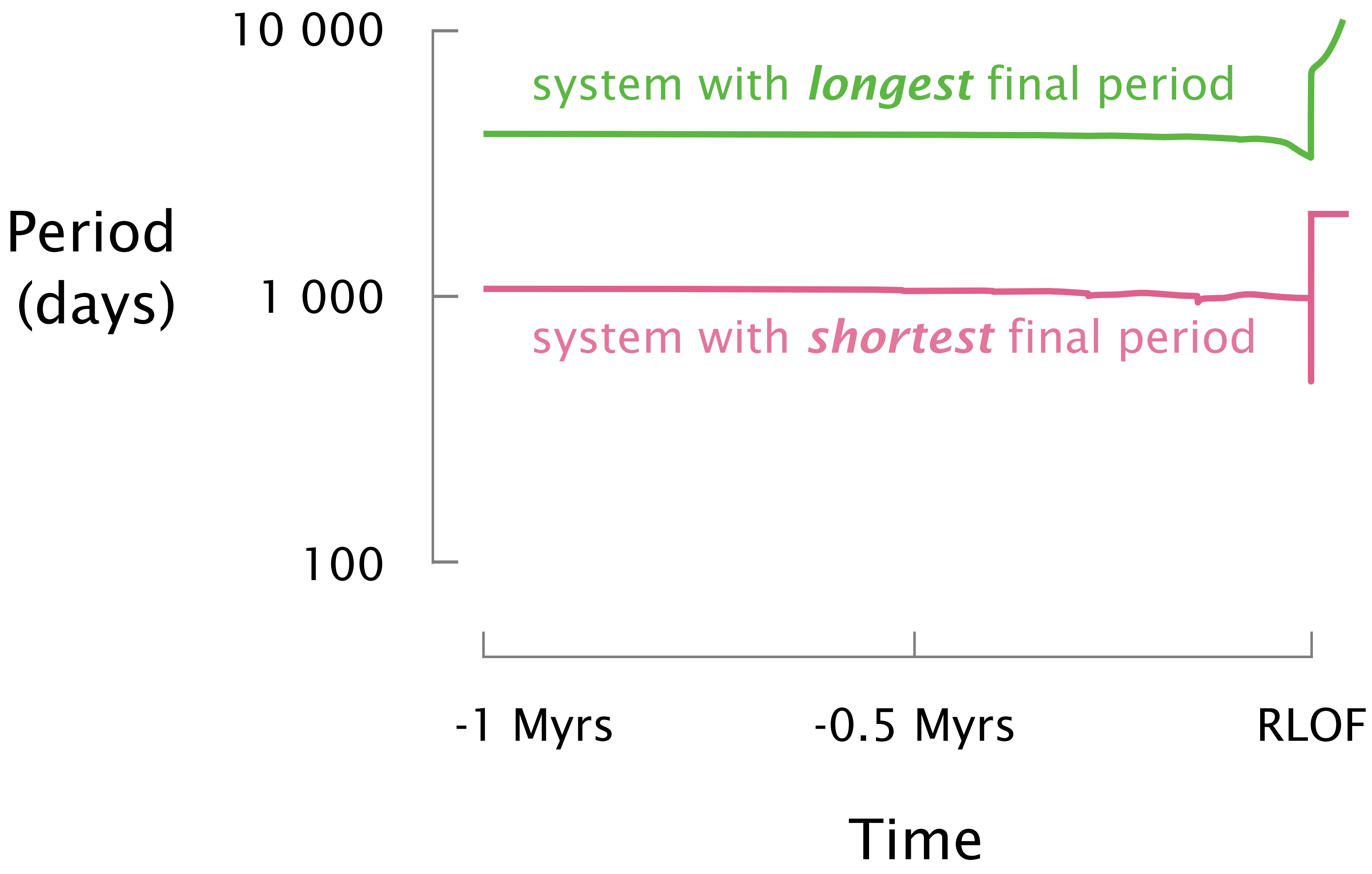


# The donor significantly overfills its Roche lobe throughout the short RLOF episode

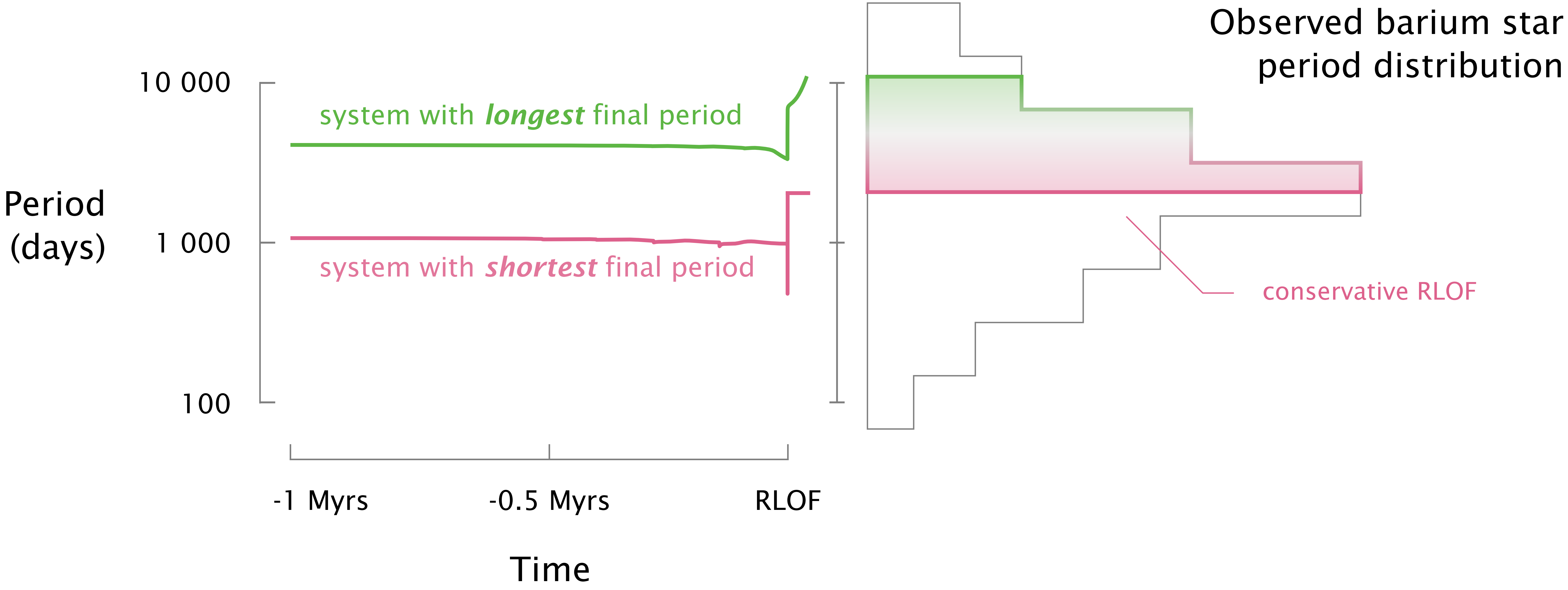


# A significant fraction of models overfill their outer Roche lobe during mass-transfer

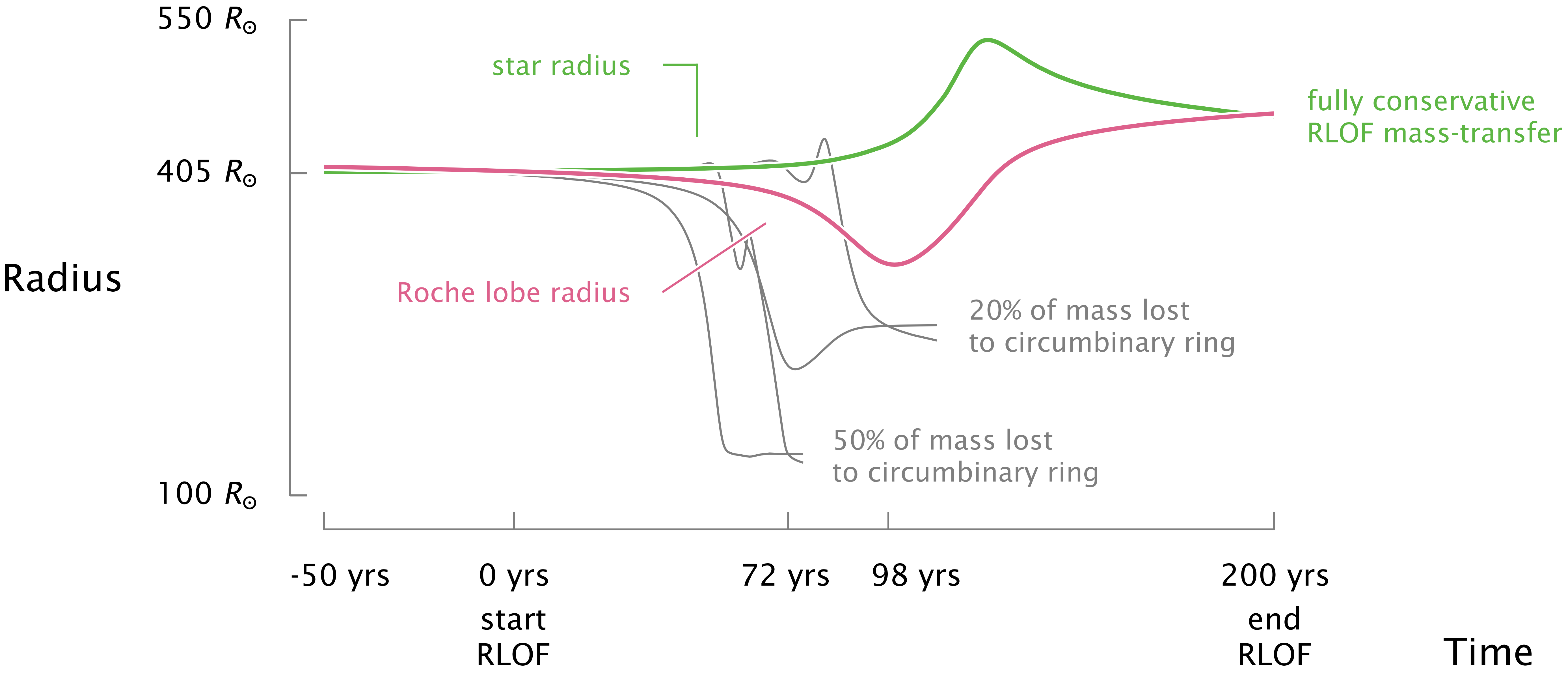



10 000
1 000
100
Period
(days)
system with ***longest*** final period
system with ***shortest*** final period
-1 Myrs
-0.5 Myrs
RLOF
Time

# Conservative RLOF mass-transfer cannot explain observed systems with short periods



# Non-conservative RLOF results in larger overfill ratios and shorter final periods



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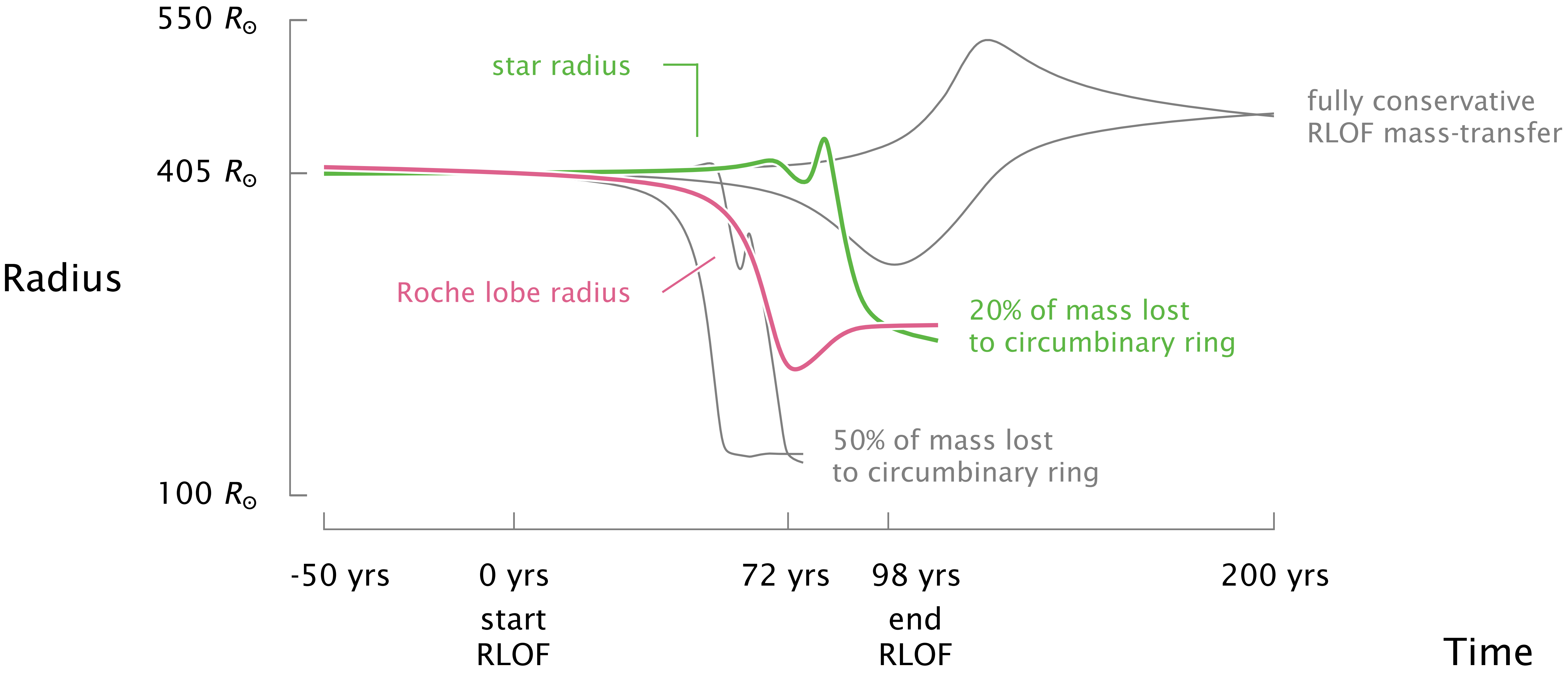

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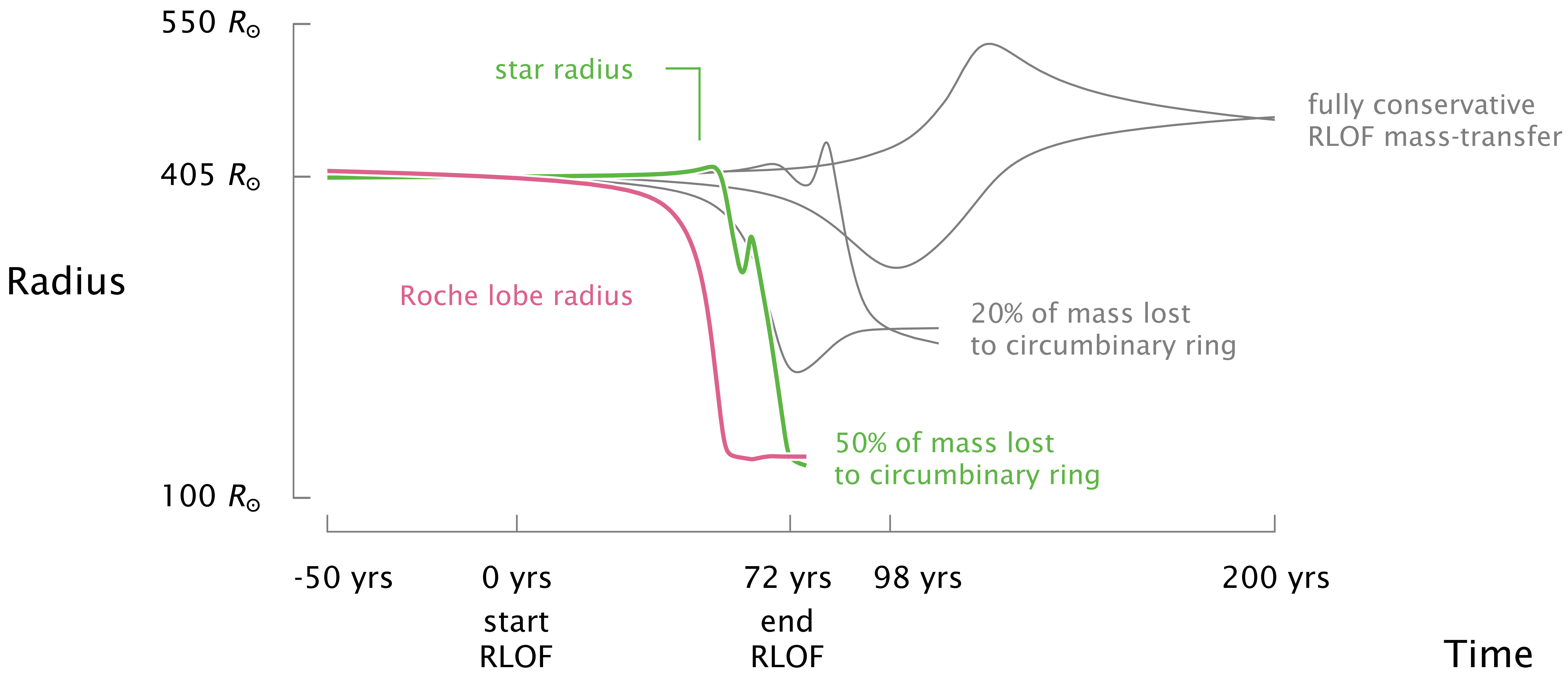
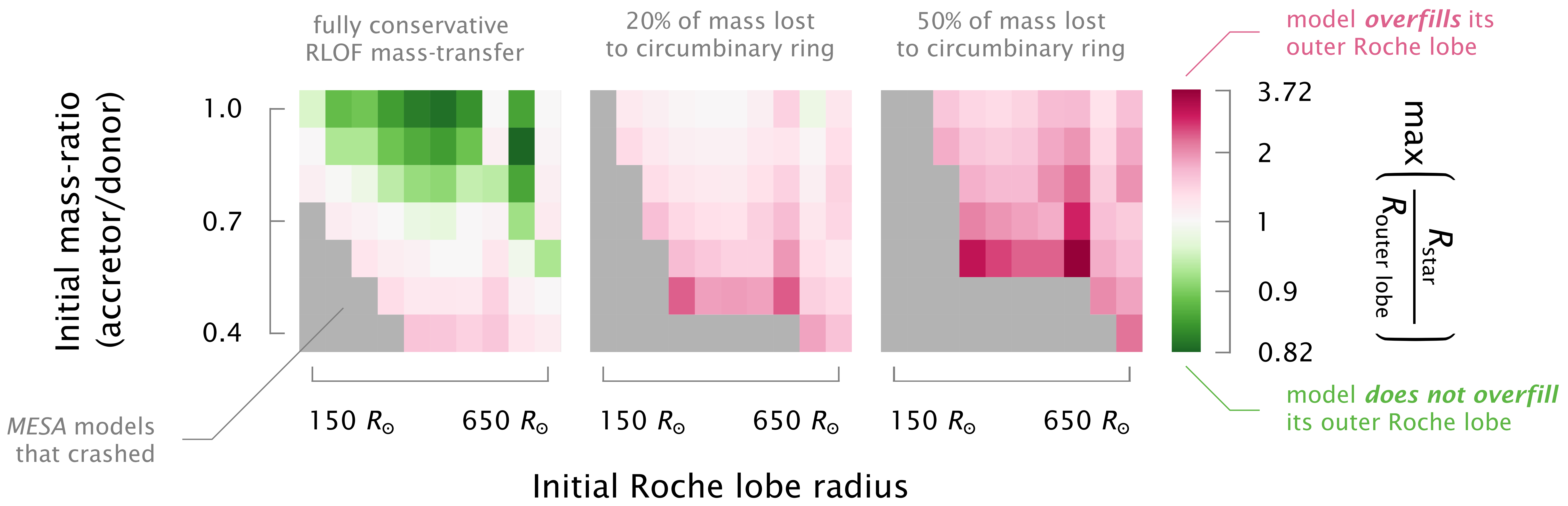
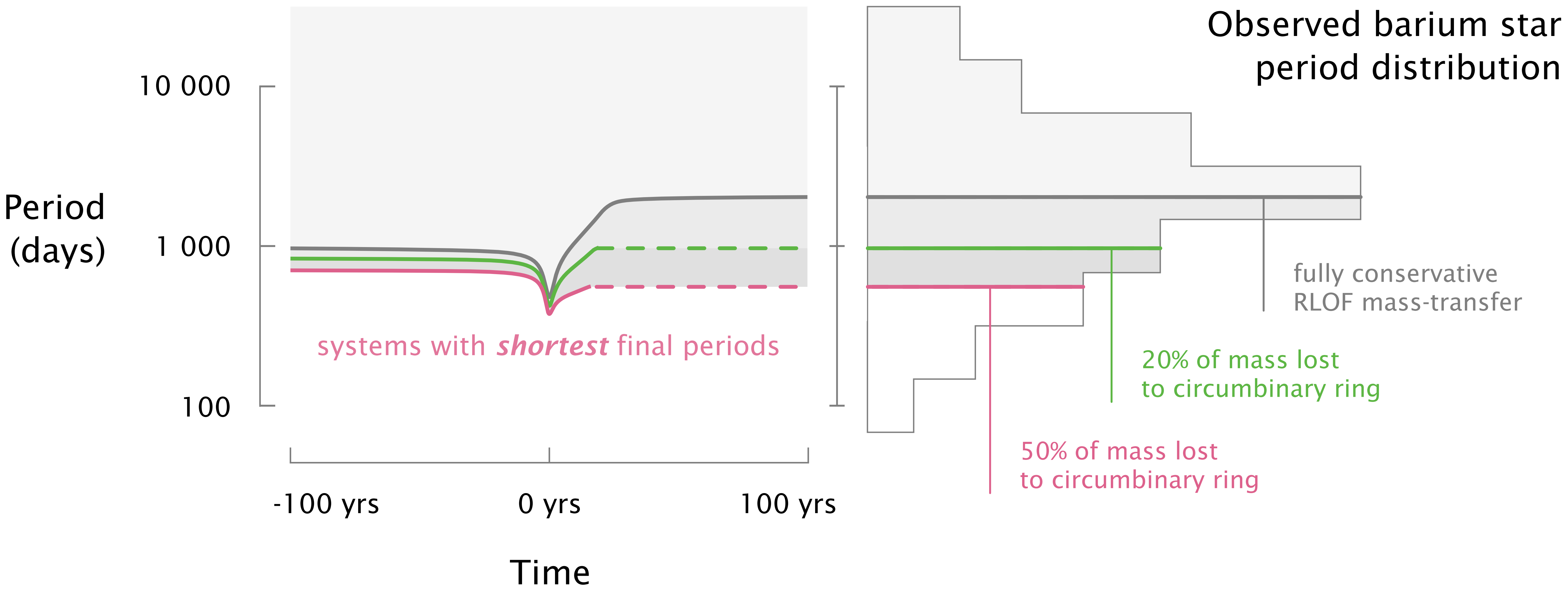


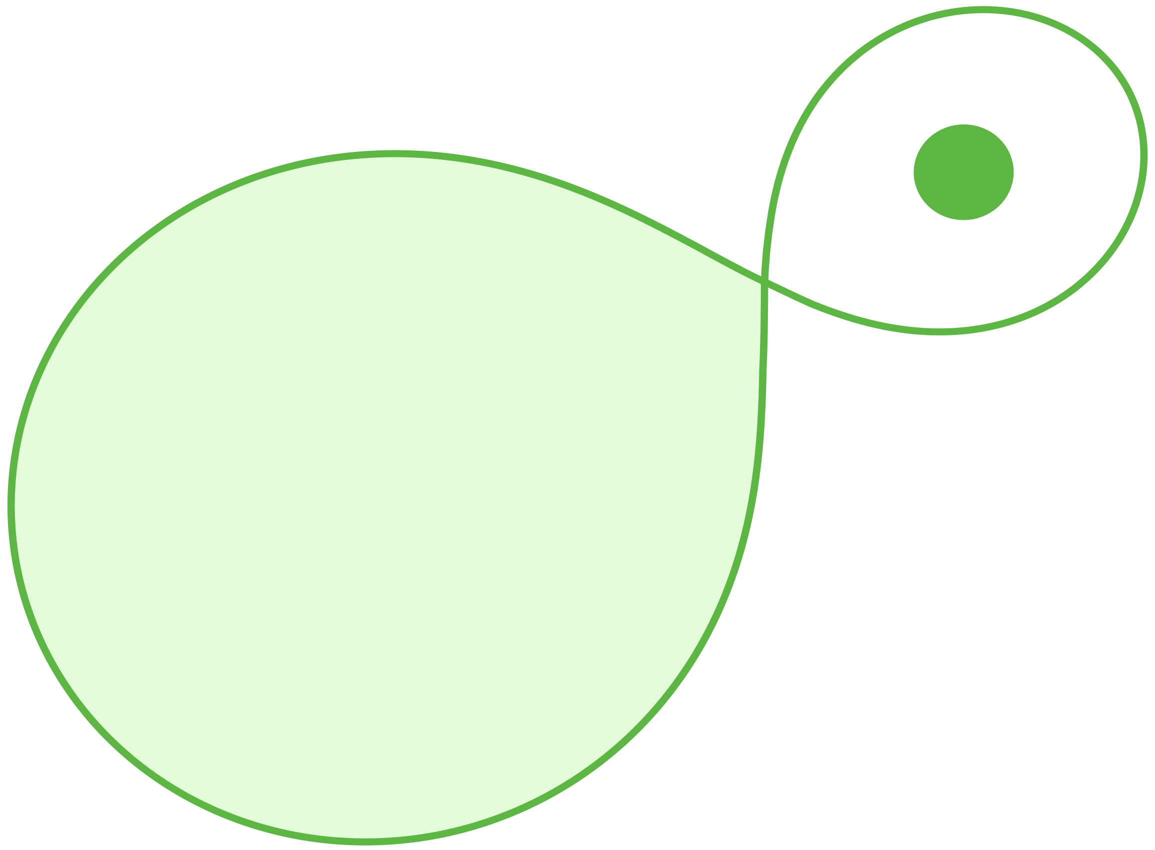
# Mass-loss to a circumbinary ring enhances outer Roche lobe overfilling



# Circumbinary mass-loss yields short periods, but still not the full observed distribution



# Forming barium stars through stable RLOF mass transfer from TPAGB-stars



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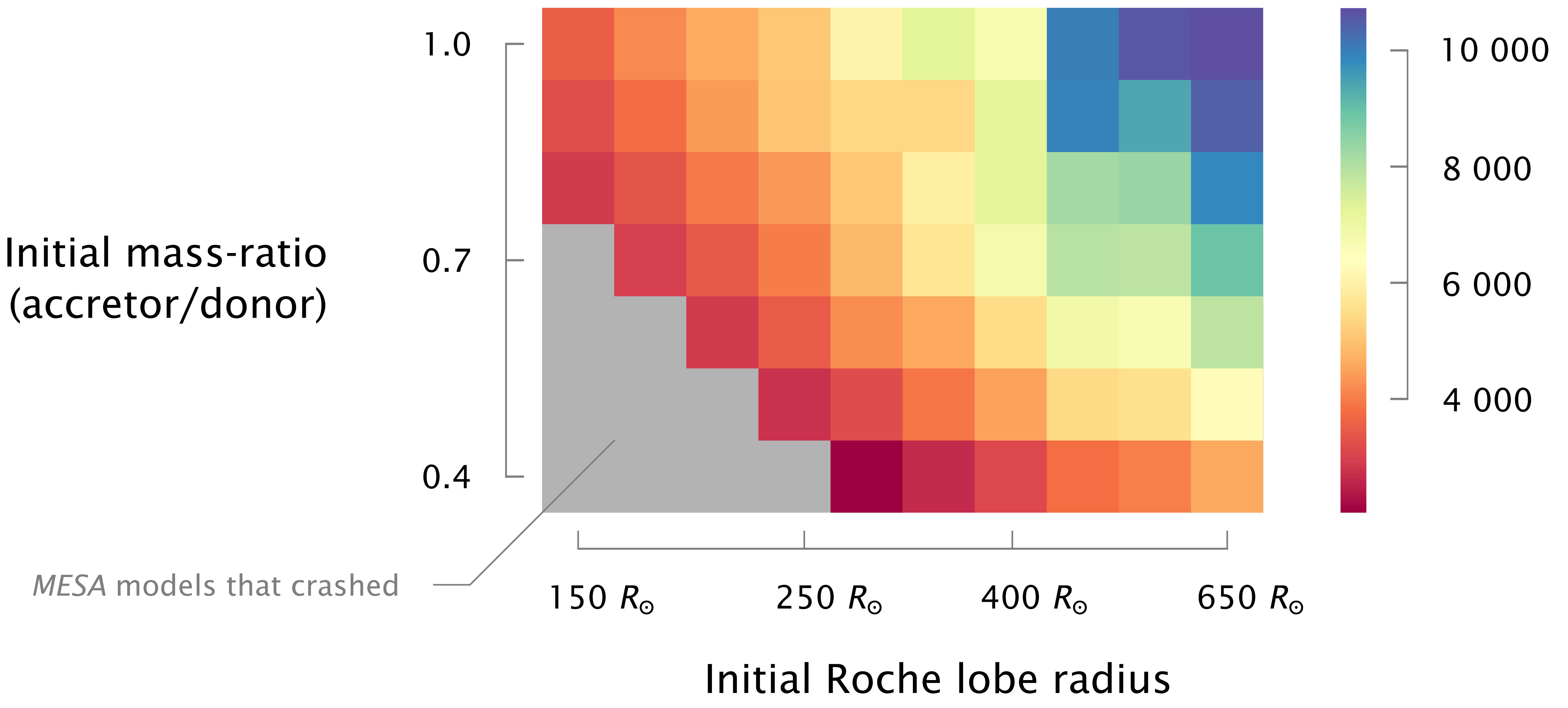
Initial mass-ratio
(accretor/donor)
1.0
0.7
0.4
MESA models that crashed
150 $R_\odot$
250 $R_\odot$
400 $R_\odot$
650 $R_\odot$
Initial Roche lobe radius
10 000
8 000
6 000
4 000
Period
(days)

# Post-mass-transfer binaries exist over a broad range of periods